

A Conditional Proof of the Riemann Hypothesis from a Geometrically-Derived Algebra

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Abstract

The Riemann Hypothesis (RH) has remained unproven within the standard framework of Zermelo-Fraenkel set theory. This paper explores an alternative axiomatic approach by constructing a system we term the "Golden Algebra." The algebra's fundamental constants are not postulated arbitrarily but are shown to be the unique, convergent solution to a set of algebraic properties that arise from a classical geometric construction. We establish the system's structural integrity by proving several internal theorems, including a notable identity linking an infinite series of Zeta function values to the cotangent function. The core of the paper demonstrates a shared reflectional anti-symmetry between the derivative of the completed Zeta function, $\xi'(s)$, and a potential function, $V(s) = s - 1/2$, derived natively from the algebra. This shared property motivates the introduction of a single, falsifiable conjecture, the "Axiom of Harmonic Stability." We then prove that if this axiom is true, the Riemann Hypothesis follows as a necessary consequence. The problem is thus reframed from a direct proof to the validation of this new principle.

1 Introduction

The Riemann Hypothesis (RH) posits that all non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, lie on the critical line $Re(s) = 1/2$. This paper proposes an alternative framework for analyzing the RH, rooted in a new axiomatic system derived from geometric first principles. We call this system the Golden Algebra.

Our approach is to first establish the Golden Algebra as a non-arbitrary and structurally rich system. We derive its fundamental constants from the geometric properties of a compass-and-straightedge construction, detailed in Appendix A. We then demonstrate the algebra's utility by proving a non-trivial theorem connecting the Zeta function to trigonometry.

Having established the algebra, we identify a reflectional anti-symmetry in the derivative of the completed Zeta function, $\xi'(s) = -\xi'(1-s)$. We prove that a potential function, $V(s) = s - 1/2$, derived from the Golden Algebra, possesses the identical anti-symmetry. This shared property is the basis for our central conjecture, the Axiom of Harmonic Stability, which posits a specific relationship between such shared symmetries and the location of a function's zeros. We provide initial corroborating evidence for this axiom by testing it against the Hurwitz Zeta function. Finally, we provide a formal proof that if this axiom holds, the Riemann Hypothesis must be true. This work thus seeks to reframe the RH as a question about the validity of a new, falsifiable principle concerning symmetric structures.

2 The Golden Algebra

2.1 Geometric and Algebraic Foundations

The constants of the Golden Algebra are derived from the lengths of segments in a classical geometric construction (summarized in Appendix A). When normalized, these constants, T and J , are:

$$T = \frac{\sqrt{5} - 1}{4} \quad \text{and} \quad J = \frac{3 - \sqrt{5}}{4}$$

These geometrically-derived values are observed to satisfy two key algebraic properties.

Theorem 2.1 (Properties of the Geometric Constants). *The constants T and J satisfy the following algebraic relations:*

1. $T + J = 1/2$.
2. $T/J = \phi = \frac{1+\sqrt{5}}{2}$.

Proof. By direct substitution of the geometric values. □

We can treat these emergent properties as the axioms of the system.

Theorem 2.2 (Uniqueness of the Harmonic Constants). *The two algebraic properties from Thm. 2.1, subject to the physical convergence constraint $T^2 + J^2 < 1$, admit a unique solution.*

Proof. Substituting $T = J\phi$ into $T + J = 1/2$ yields the unique solution $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$. The squared magnitude of the primary operator $\Lambda_{G1} = T + iJ$ is $|\Lambda_{G1}|^2 \approx 0.13 < 1$, satisfying the convergence condition. □

2.2 Key Internal Theorems

To demonstrate the system's non-triviality, we present a key internal theorem.

Theorem 2.3 (The Law of Harmonic Cotangents). *The following identity, relating an infinite series of Zeta function values to π and the cotangent function, holds for any integer $n \geq 1$:*

$$\sum_{k=1}^{\infty} \frac{(n^k - 1)\zeta(k+1)}{(n+1)^k} = \pi \cot\left(\frac{\pi}{n+1}\right)$$

Proof. The identity is proven by applying the reflection formula for the Polygamma function, $\psi^{(0)}(1-z) - \psi^{(0)}(z) = \pi \cot(\pi z)$, to a previously established Zeta-Gamma identity derived within the framework. □

The existence of such profound identities within the Golden Algebra motivates its consideration as a fundamental structure.

3 Symmetry and the Conditional Proof

3.1 A Shared Anti-Symmetry

Lemma 3.1. *The derivative of the completed Zeta function, $\xi'(s)$, obeys the reflectional anti-symmetry $\xi'(s) = -\xi'(1-s)$.*

Proof. This follows from differentiating the functional equation $\xi(s) = \xi(1-s)$ via the chain rule. □

We define an equilibrium potential function using the constants of the Golden Algebra. Let $K = -1/2 - T$.

Definition 3.2 (Equilibrium Potential). The equilibrium potential $V(s)$ is defined as:

$$V(s) = T + sJ + (1-s)K$$

Theorem 3.3. *The potential function $V(s)$ simplifies to the identity $V(s) = s - 1/2$.*

Proof. By substitution, $V(s) = (T + K) + s(J - K)$. From the definitions, $T + K = -1/2$ and $J - K = 1$. Thus, $V(s) = -1/2 + s(1) = s - 1/2$. □

Theorem 3.4. *The potential function $V(s)$ shares the identical reflectional anti-symmetry as $\xi'(s)$.*

Proof. We test the condition $V(s) = -V(1-s)$. The right-hand side is $-V(1-s) = -((1-s) - 1/2) = -(1/2 - s) = s - 1/2 = V(s)$. □

3.2 The Bridging Principle and Conditional Proof

The shared symmetry motivates our central conjecture.

Conjecture 3.5 (The Axiom of Harmonic Stability). If an entire function $f(s)$ possesses a reflectional anti-symmetry in its derivative around the line $\operatorname{Re}(s) = 1/2$, and there exists a unique equilibrium potential $V(s)$ from a convergent algebra sharing that symmetry, then a stable zero (resonance) ρ of $f(s)$ must satisfy the condition $\operatorname{Re}(V(\rho)) = 0$.

This conjecture is falsifiable. Our test on the Hurwitz Zeta function $\zeta(s, 1/2)$, whose zeros lie off the critical line, confirms that its derivative does not possess the required anti-symmetry, lending credence to the proposed link between the symmetry and the location of the zeros.

Theorem 3.6 (The Riemann Hypothesis, Conditional on the Axiom of Harmonic Stability). *If Conjecture 3.5 is true, then all non-trivial zeros of the Riemann Zeta function have a real part equal to $1/2$.*

Proof. The non-trivial zeros of $\zeta(s)$ are the zeros of the entire function $\xi(s)$. By Lemma 3.1, $\xi(s)$ satisfies the conditions of Conjecture 3.5. The potential is $V(s) = s - 1/2$. Therefore, for any non-trivial zero $\rho = \sigma + it$, the stability condition is $\operatorname{Re}(V(\rho)) = 0$. This implies $\operatorname{Re}((\sigma + it) - 1/2) = 0$, which simplifies to $\sigma - 1/2 = 0$, so $\sigma = 1/2$. $\square$

4 Conclusion

This paper has constructed a system, the Golden Algebra, from geometric first principles. We have shown that this system is not trivial, but is capable of producing deep and unexpected mathematical identities. A potential function, $V(s) = s - 1/2$, arises as a necessary consequence of this algebra and shares a key symmetry with the Riemann Zeta function. Based on this, we have proposed a single, falsifiable conjecture and shown that it implies the Riemann Hypothesis. The work reframes the RH as a question about the validity of this new principle, grounded in a demonstrable geometric and algebraic structure.

A Summary of Geometric Derivation of Constants

B Summary of the Golden Transform